

INFO 5652

Statistical Programming in R

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Electives	AI Engineer (61.5 / 100)	51.7 / 100
SPECIALIZATION On-Campus Elective		

52

CURRENT AVG SCORE
across 6 job roles

93

POTENTIAL SCORE
with recommended additions

+56

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This intensive course covers foundational data science tools and techniques in the R programming language, including acquiring, cleaning, exploring, and analyzing data, programming, and conducting reproducible research. The course will emphasize the use of data management best practices such as the tidyverse toolkit in R.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	61.5	100.0	+38.5	HIGH
Machine Learning Engineer	59.1	100.0	+40.9	HIGH
Data Scientist	53.6	100.0	+46.4	HIGH
ML Engineer	51.4	85.5	+34.1	HIGH
Applied Scientist	43.6	100.0	+56.4	HIGH
MLOps Engineer	41.1	71.8	+30.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 HIGH Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 LOW Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer

61.5 → 100.0 (+38.5 pts)

following learning software technical model systems problems machine

Machine Learning Engineer

59.1 → 100.0 (+40.9 pts)

predictive sets learning analytics solutions business mobile design

Data Scientist

53.6 → 100.0 (+46.4 pts)

sets analytics different analytical social problems

ML Engineer

51.4 → 85.5 (+34.1 pts)

discovery learning machine users systems model world problems

Applied Scientist

43.6 → 100.0 (+56.4 pts)

lab discovery scientific computational reasoning literature problems technical

MLOps Engineer

41.1 → 71.8 (+30.7 pts)

model systems learning machine development error

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.